

VARIABLE STARS

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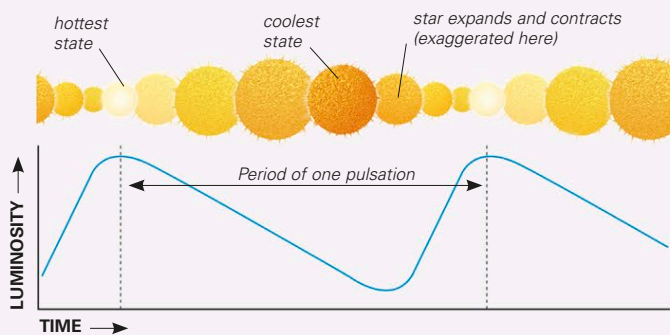
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ALTHOUGH AT FIRST SIGHT the stars in the night sky seem to be unchanging, many thousands of stars change their brightness over periods ranging from a few days to decades. True, or intrinsic, variable stars vary in brightness due to physical changes within the star. Others, such as eclipsing binaries (see p.274), only appear to vary because they have orbiting companions.

PROTOTYPE

Mira is one of the most famous variable stars in the Milky Way (see p.285). It is a long-period, pulsating star that has given its name to one of the main types of variable stars.



PULSATING VARIABLES

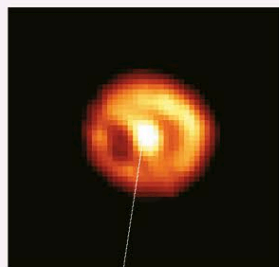
Pulsating variable stars are intrinsic variables that undergo repetitive expansion and contraction of their outer layers. A pulsating star is constantly trying to reach equilibrium between the inward gravitational force and the outward radiation and gas pressure. This causes the star's brightness to vary. In many types of pulsating stars, including Cepheids (see p.286), a star's period of variation is related to its luminosity. Knowledge of the star's luminosity, coupled with its apparent magnitude, enables astronomers to calculate a star's distance. Pulsating variables are therefore a useful tool for determining distances to faraway objects such as other galaxies.

LIGHT CURVE

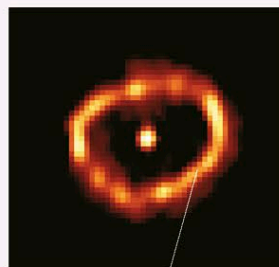
The light curve of a Cepheid variable shows the regular variation in luminosity during a period of one pulsation.

NOVAE

A nova is a binary system consisting of a giant star that is being orbited by a smaller white dwarf. The giant star has grown so large that its outer material is no longer gravitationally bound to the star and instead falls onto the white-dwarf companion. Eventually, this gain in material triggers a thermonuclear explosion on the surface of the white dwarf, which brightens it by many magnitudes, increasing its energy output by a factor of a million or more. The surface gases of the white dwarf are in a "degenerate" state, and they do not obey the normal gas laws. Usually a gas explosion will cause the gas to expand, thereby reducing the explosion and finishing it. However, the degenerate gases of a white dwarf do not expand, and the explosion turns into a runaway event that does not finish until the fuel is exhausted. Prior to this, the binary system would be invisible to the naked eye, and the nova outburst would then bring the system into visibility as a "new"—in Latin, *nova*—star.



exploded star



hot bubble of gas

CATAclysmic BINARY

The most widely studied nova, Nova Cygni 1992, was witnessed exploding in 1992 (see p.287). Its magnitude rose by such a degree that at its brightest the nova was visible to the naked eye.

LIGHT ECHOES

In 2002, the star V838 Monocerotis (see p.265) emitted an outburst of light that echoed off the surrounding dust. The star has been imaged several times since. At first thought to be a nova, it might in fact be a new type of eruptive star.

MAY 20, 2002

SEPTEMBER 2, 2002

OCTOBER 28, 2002